

String = It is a sequence of character enclosed in single, double, or triple quotes

- string is immutable

```
In [1]: x = "    MY name is md yasir arafat"
print(x)
```

MY name is md yasir arafat

```
In [2]: print(x[0])
```

```
In [3]: print(x[0:5])
```

M

```
In [4]: print(x[0:len(x)])
```

MY name is md yasir arafat

```
In [5]: print(x[0:len(x):2])
```

M aei dysraaa

```
In [7]: print(x[0:5:2])
```

M

```
In [8]: print(x[::2])
```

M aei dysraaa

```
In [9]: x = "yasir"
print(x[::-1])
```

risay

```
In [10]: print(x.lower())
```

yasir

```
In [11]: print(x.upper())
```

YASIR

```
In [12]: print(x.strip())
```

yasir

```
In [13]: print(x.replace("y", "a"))
```

aasir

```
In [14]: print(x.count("a"))
```

1

```
In [15]: print(x.startswith("m"))
```

False

```
In [16]: print(x.endswith("t"))
```

False

```
In [17]: print(x.capitalize())
```

Yasir

```
In [18]: print(x.find("y"))
```

0

```
In [19]: x = 'yasir arafat'
```

```
print(x.split())
```

['yasir', 'arafat']

```
In [20]: x = "Yasir"
```

```
print(x.swapcase())
```

yASIR

```
In [21]: x = 'yasir arafat'
```

```
y = x.split()
```

```
print(max(y, key=len))
```

arafat

```
In [22]: x = 'yasir1232'
```

```
print(x.isalpha()) # this function is check wether the input is char or n
```

False

```
In [23]: #Check if a string contains only digits.
```

```
x = "12334"
```

```
if x.isdigit():
```

```
    print("the string contains only digits")
```

```
else:
```

```
    print("this sting does not contain only digits")
```

the string contains only digits

```
In [24]: #Check if the string is empty or not.
```

```
str = ""
```

```
if str:
```

```
    print("string is not emempty")
```

```
else:
```

```
    print("string is empty")
```

string is empty

```
In [25]: #Remove all spaces from a string.
```

```
name = "md yasir arafat"
```

```
x = name.replace(" ", "")
```

```
print(x)
```

mdyasirarafat

In [26]: *#Check if a string is a palindrome (same forward and backward).*

```
text = "12321"

if text==text[::-1]:
    print("this is palandrome")
else:
    print("this is not a paladrome")
```

this is palandrome

In [27]: *#Count the number of vowels and consonants in a string.*

```
name = "Yasir"

name_lower = name.lower()
vowel = "aeiou"
vowel_count = 0
consonant_count = 0

for char in name_lower:
    if char.isalpha():
        if char in vowel:
            vowel_count +=1
        else:
            consonant_count +=1
print(f"vowels :{vowel_count}")
print(f"consonants :{consonant_count}")
```

vowels :2

consonants :3

In [28]: *# Replace all occurrences of "a" with "#" in a string.*

```
x = 'yasir arafat'

print(x.replace("a","#"))
```

y#sir #r#f#t

In [29]: *# Remove all punctuation from a string*

```
import string # this module contain useful function for string

x = "my, n:am;e, is, md, yasir, ar;afat"

no_punct = "".join(ch for ch in x if ch not in string.punctuation) #str
cleaned = " ".join(no_punct.split())
print(no_punct)
```

my name is md yasir arafat

In [30]: *# Split a string into words and print each word on a new line.*

```
x = 'yasir'
```

```
y = x.split()

for y in x:
    print(y)
```

y
a
s
i
r

In [31]: *# Join a list of words into a single string separated by " - ".*

```
x = ['md', 'yasir', 'arafat']

result = "-".join(x)
print(result)
```

md-yasir-arafat

In [32]: *# Swap the case of each character in a string.*

```
x = 'Yasri Arafat'

y = x.swapcase()
print(y)
```

yASRI aRAFAT

In [33]: *# Remove duplicate characters from a string.*

```
x = "yasir arafat"

result = ""

for ch in x:
    if ch not in result:
        result += ch

print(result)
```

yasir ft

In [34]: *# Find the longest word in a sentence.*

```
x = 'md yasir arafat'

words = x.split()

longest = max(words, key=len)

print('longest word is :', longest)
```

longest word is : arafat

In [35]: *# Count how many words in a sentence start with a vowel.*

```
x = 'md yasir arafta'

x_lower = x.lower()
```

```

vowel = 'aeiou'

start_with_vowel = 0

x_split = x_lower.split()

print(x_split)

for char in x_split:
    if char[0] in vowel:
        start_with_vowel +=1

print(start_with_vowel)

```

```

['md', 'yasir', 'arafta']
1

```

In [36]: *# Reverse the order of words in a sentence without using reverse() functi*

```

x = 'yasir arafat'
print(x.split())
print(x.split()[::-1])

```

```

['yasir', 'arafat']
['arafat', 'yasir']

```

In [37]: *# Check if a string contains all letters of the English alphabet (pangram*